Team 29 MGT 6203 Project Proposal

Team Members:

1. Olivia Watson (owatson6): Olivia graduated with a degree in Marketing and Spanish from the University of Miami. She now works as a senior analyst at New Balance.
2. Selda Kocaman (skocaman7): Selda earned her bachelor's degree in economics from Marmara University of Turkey and works as a Business Systems Analyst II at Dept of Health and Human Service in NC.
3. Kelly Cristina Ribeiro Yogui (kyogui3): Kelly graduated in Applied Mathematics from the University of São Paulo (Brazil). She is currently working as Data Specialist supporting diverse business teams.
4. Thu Thi Diem Le (tdiem6): Thu earned an MBA and a BSc. in Chemistry from National University of Singapore. Thu is currently working as a Data Analyst at the Institute for Community Alliances.
5. Hande Pehlivan (hpehlivan3): Hande graduated with a master's degree in biomedical engineering. Her most recent projects include heart disease prediction and stock analyses application.

Project Title:

Diabetes probability based on behavioral, medical, demographic and economic factors

Background Information & Problem Statement:

Nearly 10% of Americans have type II diabetes (CDC), costing each nearly 10 years of their life, and the U.S. over $325 billion dollars, annually. The American Diabetes Association (ADA) states that "*Care for people with diagnosed diabetes accounts for one in four health care dollars in the U.S.*" We seek to identify behavioral and societal diabetes risk factors to improve health care programs, lower the U.S.'s healthcare burden, improve the private sector's productivity, and improve U.S. citizens' wellbeing.

Primary & Secondary Questions:

Primary: What factors impact a person's probability of having diabetes?
How strong is the relationship between diabetes and other diseases (obesity, heart disease, etc.)?
How does diet and exercise impact the probability of having a diabetes diagnosis?
How does cost of living impact the probability of having a diabetes diagnosis?

Business Justification:

The ADA declares that "*The total estimated 2017 cost of diagnosed diabetes of $327 billion includes $237 billion in direct medical costs and $90 billion in reduced productivity. […] People with diagnosed diabetes incur average medical expenditures of $16,752 per year, of which about $9,601 is attributed to diabetes. On average, people with diagnosed diabetes have medical expenditures approximately 2.3 times higher than what expenditures would be in the absence of diabetes.*"
Understanding the main factors which may lead to a diabetes diagnosis is paramount to our society's survival. For citizens, this means extending their life expectancy, avoiding nearly $10k

in diabetes associated costs per year, and enhancing their health. The private sector could save $90 billion dollars from improved productivity, reduced absenteeism, and reduced insurance costs. Governments could re-allocate ¼ of all health care dollars to solving other crises or reduce taxes to reflect reduced health care needs.

Datasets:

1. CDC - 2021 BRFSS Survey Data and Documentation.
   BRFSS dataset includes behavioral, medical condition, and demographic data. More on the background of the BRFSS survey can be found here. See Appendix fig. 1 screenshot.
2. Cost-of-living index
   The 2022 Annual Average Cost of Living table has the Cost-of-Living Index and Grocery, Housing, Utilities, Transportation, Health, and Misc. values for US states. See Appendix fig. 2 screenshot.

Key Variables:

We will set the diabetes diagnosis as the dependent variable. From our research, we expect behavioral (diet, physical exercise, ability to work), demographic (sex at birth, age, state) and medical condition (cancer, heart attack) factors may be correlated with diabetes occurrence. We will use the cost-of-living data to understand if a resident's state's economic environment, including the cost of Health, Grocery, and Housing expenses, impacts the probability of receiving a diabetes diagnosis.

Approach:

1. **Data Processing & Cleaning:** We first will decide how to treat missing data. We will then merge this dataset with the cost-of-living dataset using their common variables, state and year.
2. **Choosing the features for predictive models**: We will run QQ and correlation charts to identify which variables are highly correlated and if we need to include interaction variables. We will use feature selection methods, including ridge and stepwise to decide which features to include in the models.
3. **Run Logistic Regression, KNN Random Forest, and SVM with cross-validation. Compare the cross-validation accuracy results.** We will try other models as needed and choose the model that has the best accuracy score.
4. **Split the data into training and test sets.** Use selected model and variables from previous steps.
5. **Build Graphs and Analyze results**

Anticipated Conclusions:

We expect that behavioral and economic factors correlate with the probability of developing diabetes. We will run machine learning models to predict how likely a person is to have the disease. We anticipate that the most significant variables will be related to obesity, diet, exercise, and cost of living. Also, these fields may be correlated, leading to interaction terms.

Governments, citizens, and companies may use the results to guide palliative actions to avoid diabetes. Additionally, it may help health workers to predict possible diabetes cases to reverse it.

Timeline:

Project Task	Start Date	Due Date	Assignee to complete
Brainstorm topic, find data, speak with Akshay, create proposal	02-17	03-11	All members will be involved at all stages. Certain team members will play a lead role in different tasks according to their strengths.
Continue Researching	03-10	03-15	
Data Cleaning	03-15	03-20	
Project Plan Presentation Video	03-18	03-26	
Preliminary Data Modeling/ algorithm tuning	03-20	04-02	
Project Progress Report	03-20	04-02	
Compare, develop, and analyze models	03-25	04-10	
Create Final Report	04-01	04-10	
Create Final Report with Code	04-01	04-16	
Create Final Video Presentation	04-01	04-19	

Appendix:

Figure 1:

	X_STATE	FMONTH	IDATE	IMONTH	IDAY	IYEAR	DISPCODE	SEQNO	X_PSU	CTELENM1	PVTRESD1	COLGHOU
1	1	1	01192021	01	19	2021	1100	2021000001	2.021e+09	1	1	
2	1	1	01212021	01	21	2021	1100	2021000002	2.021e+09	1	1	
3	1	1	01212021	01	21	2021	1100	2021000003	2.021e+09	1	1	
4	1	1	01172021	01	17	2021	1100	2021000004	2.021e+09	1	1	
5	1	1	01152021	01	15	2021	1100	2021000005	2.021e+09	1	1	

Figure 2:

Rank	State	Index	Grocery	Housing	Utilities	Transportation	Health	Misc.
1	Mississippi	85.0	92.4	67.4	89.0	91.9	97.7	91.6
2	Oklahoma	85.8	93.7	70.2	95.1	90.9	91.2	90.4
3	Kansas	87.5	93.7	71.1	98.0	95.6	100.4	91.6
4	Alabama	88.1	97.6	69.6	100.7	89.9	89.6	95.0
5	Georgia	88.6	94.6	75.6	90.3	89.8	94.6	95.1